

Use containerization technology to build the postgres + pg4admin environment

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1. Install postgres database and run

```
gitpod /workspace/empty (main) $ mkdir ./pg_files
```

```
gitpod /workspace/empty (main) $ docker pull postgres
```

```
gitpod /workspace/empty (main) $ docker ps
```

```
gitpod /workspace/empty (main) $ docker images
```

```
gitpod /workspace/empty (main) $ docker run --name postgres_container -e
POSTGRES_PASSWORD=12345 -d -p 5432:5432 -v
/workspace/empty/pg_files:/var/lib/postgresql/data postgres
```

```
gitpod /workspace/empty (main) $
```

```
gitpod /workspace/empty (main) $ docker exec -it postgres_container psql -U postgres
```

```
psql (17.5 (Debian 17.5-1.pgdg120+1))
```

Type "help" for help.

```
postgres=# CREATE DATABASE flight;
```

```
CREATE DATABASE
```

```
postgres=# \q
```

```
postgres=# \l
```

List of databases								
Name	Owner	Encoding	Locale Provider	Collate	Ctype	Locale	ICU Rules	Access privileges
flight	postgres	UTF8	libc	en_US.utf8	en_US.utf8			
postgres	postgres	UTF8	libc	en_US.utf8	en_US.utf8			
template0	postgres	UTF8	libc	en_US.utf8	en_US.utf8			=c/postgres +
								postgres=CTc/postgres
template1	postgres	UTF8	libc	en_US.utf8	en_US.utf8			=c/postgres +
								postgres=CTc/postgres

(4 rows)

```
postgres=# █
```

Screenshot

```
postgres=# \c flight
You are now connected to database "flight" as user "postgres".
```

```
flight=# CREATE TABLE faa_airport_stats (
    id INTEGER PRIMARY KEY,
    passenger_count INTEGER,
    Quarter TEXT,
    Avg_Comp_Airfare NUMERIC(6,2),
    GDP NUMERIC(12,2),
    Q1 NUMERIC(6,2),
    Q2 NUMERIC(6,2),
    Q3 NUMERIC(6,2),
    Q_Y NUMERIC(6,2),
    COVID_19_ TEXT,
    Net_Income_Fin_Position_ NUMERIC(10,2),
    Weather_Natural_Disaster TEXT,
    Avg_Airfare NUMERIC(6,2),
    Year INTEGER,
    Entry_Exit_in_Market TEXT,
    Recession BOOLEAN,
    Price_Alt_Transport_Gasoline_ NUMERIC(6,2)
);
CREATE TABLE
flight=# \dt
public | faa_airport_stats | table | postgres
```

```
CREATE TABLE faa_airport_stats (
    id INTEGER PRIMARY KEY,
    passenger_count INTEGER,
    Quarter TEXT,
    Avg_Comp_Airfare NUMERIC(6,2),
    GDP NUMERIC(12,2),
    Q1 NUMERIC(6,2),
    Q2 NUMERIC(6,2),
    Q3 NUMERIC(6,2),
    Q_Y NUMERIC(6,2),
    COVID_19_ TEXT,
    Net_Income_Fin_Position_ NUMERIC(10,2),
    Weather_Natural_Disaster TEXT,
    Avg_Airfare NUMERIC(6,2),
    Year INTEGER,
    Entry_Exit_in_Market TEXT,
    Recession BOOLEAN,
    Price_Alt_Transport_Gasoline_ NUMERIC(6,2)
);
```

2. Create database in postgresQL server container and create the database and tables

1.1 Import SQL into gitpod's postgres:

- Drag and drop 'flights-export.csv' in gitpod's workspace:

- Install postgresql-client:
 - `sudo apt update`
 - `sudo apt install postgresql-client`
- Login to SQL CLI and create a database
 - `psql -h localhost -p 5432 -U postgres`
OR
`docker exec -it postgres_container psql -U postgres`
`CREATE DATABASE flight;`
`CREATE TABLE faa_airport_stats (`
`id INTEGER PRIMARY KEY,`
`passenger_count INTEGER,`
`Quarter TEXT,`
`Avg_Comp_Airfare NUMERIC(6,2),`
`GDP NUMERIC(12,2),`
`Q1 NUMERIC(6,2),`
`Q2 NUMERIC(6,2),`
`Q3 NUMERIC(6,2),`
`Q_Y NUMERIC(6,2),`
`COVID_19_ TEXT,`
`Net_Income_Fin_Position_ NUMERIC(10,2),`
`Weather_Natural_Disaster TEXT,`
`Avg_Airfare NUMERIC(6,2),`
`Year INTEGER,`
`Entry_Exit_in_Market TEXT,`
`Recession BOOLEAN,`
`Price_Alt_Transport_Gasoline_ NUMERIC(6,2)`

```
);  
\\q
```

○ Import the sql

```
psql -U postgres -d flight -h localhost -c "\copy faa_airport_stats FROM  
'flights-export.csv' WITH (FORMAT csv, HEADER true, DELIMITER ',')"
```

```
● gitpod /workspace/empty (main) $ sudo apt install postgresql-client  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  postgresql-client-14 postgresql-client-common  
Suggested packages:  
  postgresql-14 postgresql-doc-14  
The following NEW packages will be installed:  
  postgresql-client postgresql-client-14 postgresql-client-common  
0 upgraded, 3 newly installed, 0 to remove and 25 not upgraded.  
Need to get 1,279 kB of archives.  
After this operation, 4,427 kB of additional disk space will be used.  
Do you want to continue? [Y/n] y  
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 postgresql-client-common all 238 [29.6 kB]  
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 postgresql-client-14 amd64 14.18-0ubuntu0.22.04.1 [1,246 kB]  
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 postgresql-client all 14+238 [3,292 B]  
Fetched 1,279 kB in 1s (967 kB/s)  
debconf: delaying package configuration, since apt-utils is not installed  
Selecting previously unselected package postgresql-client-common.  
(Reading database ... 36571 files and directories currently installed.)  
Preparing to unpack .../postgresql-client-common_238_all.deb ...  
Unpacking postgresql-client-common (238) ...  
Selecting previously unselected package postgresql-client-14.  
Preparing to unpack .../postgresql-client-14_14.18-0ubuntu0.22.04.1_amd64.deb ...  
Unpacking postgresql-client-14 (14.18-0ubuntu0.22.04.1) ...  
Selecting previously unselected package postgresql-client.  
Preparing to unpack .../postgresql-client_14+238_all.deb ...  
Unpacking postgresql-client (14+238) ...  
Setting up postgresql-client-common (238) ...  
Setting up postgresql-client-14 (14.18-0ubuntu0.22.04.1) ...  
update-alternatives: using /usr/share/postgresql/14/man/man1/psql.1.gz to provide /usr/share/man/man1/psql.1.gz (psql.1.gz) in auto mode  
Setting up postgresql-client (14+238) ...  
Processing triggers for man-db (2.10.2-1) ...  
○ gitpod /workspace/empty (main) $  
○ gitpod /workspace/empty (main) $  
● gitpod /workspace/empty (main) $ psql -U postgres -d flight -h localhost -c "\copy faa_airport_stats FROM 'flights-export.csv' WITH (FORMAT csv, HEADER true, DELIMITER  
Password for user postgres:  
psql: error: connection to server at "localhost" (127.0.0.1), port 5432 failed: fe_sendauth: no password supplied  
● gitpod /workspace/empty (main) $ psql -U postgres -d flight -h localhost -c "\copy faa_airport_stats FROM 'flights-export.csv' WITH (FORMAT csv, HEADER true, DELIMITER  
Password for user postgres:  
COPY 66
```

```
○ gitpod /workspace/empty (main) $ docker exec -it postgres_container psql -U postgres  
psql (17.5 (Debian 17.5-1.pgdg120+1))  
Type "help" for help.
```

```
postgres=# \l
```

Name	Owner	Encoding	Locale Provider	List of databases		Locale	ICU Rules	Access privileges
				Collate	Ctype			
flight	postgres	UTF8	libc	en_US.utf8	en_US.utf8			
postgres	postgres	UTF8	libc	en_US.utf8	en_US.utf8			
template0	postgres	UTF8	libc	en_US.utf8	en_US.utf8			=c/postgres +
template1	postgres	UTF8	libc	en_US.utf8	en_US.utf8			postgres=Ctc/postgres +
								=c/postgres +
								postgres=Ctc/postgres

(4 rows)

```
postgres=# \c flight  
You are now connected to database "flight" as user "postgres".  
flight=# \dt
```

List of relations			
Schema	Name	Type	Owner
public	faa_airport_stats	table	postgres

(1 row)

```
flight=# select * from faa_airport_stats;
```

Another Option is to import data using PgAdmin

3. install pgadmin4 and run it

gitpod /workspace/empty (main) \$ docker pull dpage/pgadmin4

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● gitpod /workspace/empty (main) \$ docker ps

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
6fb021b69db6	postgres	"docker-entrypoint.s..."	46 minutes ago	Up 46 minutes	0.0.0.0:5432->5432/tcp

postgres_container

○ gitpod /workspace/empty (main) \$

● gitpod /workspace/empty (main) \$ docker pull dpage/pgadmin4

Using default tag: latest

latest: Pulling from dpage/pgadmin4

fe07684b16b8: Pull complete

cd561da15499: Pull complete

09e0515c2c45: Pull complete

96d90848fc93: Pull complete

a205e07afe69: Pull complete

76d41f9442f9: Pull complete

a7301d9d96df: Pull complete

710108fc044a: Pull complete

bc9209d2a2d8: Pull complete

3d99b9e2f3dd: Pull complete

10498d72e7aa: Pull complete

fd49a04e2cb7: Pull complete

11ae003a502d: Pull complete

8900417a7546: Pull complete

14c04d8b18eb: Pull complete

Digest: sha256:2a830466aafd9cbc2aea0f76ff81a80dbbba819f2f5db7e69cb40e9cbdb6bc7b

Status: Downloaded newer image for dpage/pgadmin4:latest

docker.io/dpage/pgadmin4:latest

○ gitpod /workspace/empty (main) \$ █

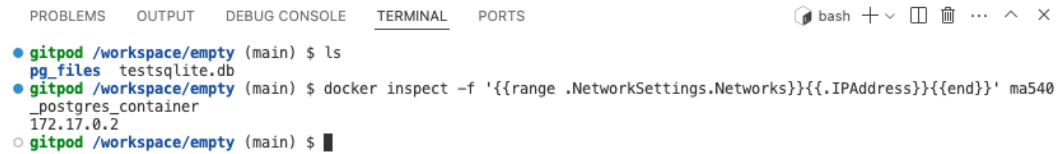
gitpod /workspace/empty (main) \$ docker run --name pgadmin_container -p 8080:80 -e 'PGADMIN_DEFAULT_EMAIL=im.bidur@gmail.com' -e 'PGADMIN_DEFAULT_PASSWORD=12345' -d dpage/pgadmin4

Port	Address	Protocol	Description	State
5432	https://5432-gitpodio-empty-10s8yiot1lp.ws-us120.gitpod.io	HTTP		open (public)
8080	https://8080-gitpodio-empty-10s8yiot1lp.ws-us120.gitpod.io	HTTP		open (public)
37405	https://37405-gitpodio-empty-10s8yiot1lp.ws-us120.gitpod.io	HTTP		open (private)

Make sure that the Port for pgadmin4 is **public**.

Note the IP address of the postgres container

gitpod /workspace/empty (main) \$ docker inspect -f
'{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}' postgres_container



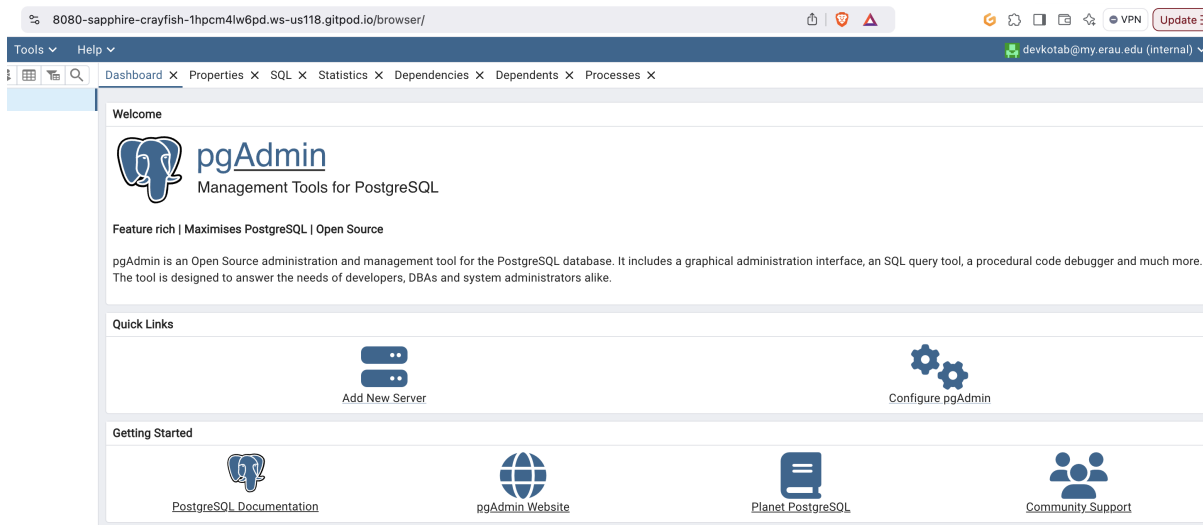
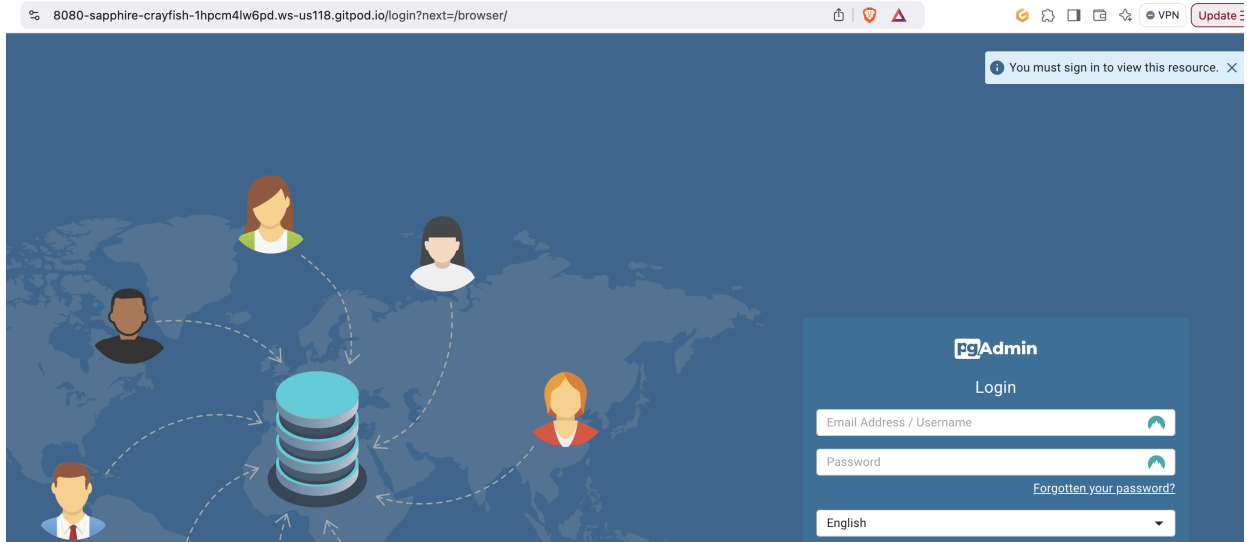
The screenshot shows a terminal window with tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The terminal shows the following commands and output:

```
gitpod /workspace/empty (main) $ ls
pg_files  testsqlite.db
gitpod /workspace/empty (main) $ docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}' postgres_container
172.17.0.2
gitpod /workspace/empty (main) $
```

pgAdmin is located at: <https://8080-gitpodio-empty-10s8yiot1lp.ws-us120.gitpod.io>

4. Access PgAdmin

- Open PgAdmin in browser: <https://8080-sapphire-crayfish-1hpcm4lw6pd.ws-us118.gitpod.io/login?next=/>
- Login to PgAdmin: Provide the login credentials specified at environment variables <PGADMIN_DEFAULT_EMAIL> and <PGADMIN_DEFAULT_PASSWORD>:
 - Email: im.bidur@gmail.com
 - Password: 12345



- **Create a New Server Connection**
 - In the **Dashboard**, click on **"Add New Server"**.
 - In the **"General"** tab:
 - **Name:** DS500-POSTGRES

Register - Server

GeneralConnectionParametersSSH TunnelAdvancedPost Connection SQLTags

Name

bd_postgres

Server group

Servers

Background

X

Foreground

X

Connect now?

Shared?

Shared Username

Comments

Either Host name or Service must be specified.

i?

X Close

Reset

Save

- In the "Connection" tab, enter the necessary fields as shown in the screenshot and save.

The screenshot shows a 'Register - Server' dialog box with the 'Connection' tab selected. The dialog has a title bar with a close button (X) and a tab bar with 'General', 'Connection', 'Parameters', 'SSH Tunnel', 'Advanced', and 'Tags'. The 'Connection' tab contains the following fields and controls:

- Host name/address: 172.17.0.2
- Port: 5432
- Maintenance database: postgres
- Username: postgres
- Kerberos authentication?: ☐
- Password:
- Save password?: ☒
- Role: (empty field)
- Service: (empty field)

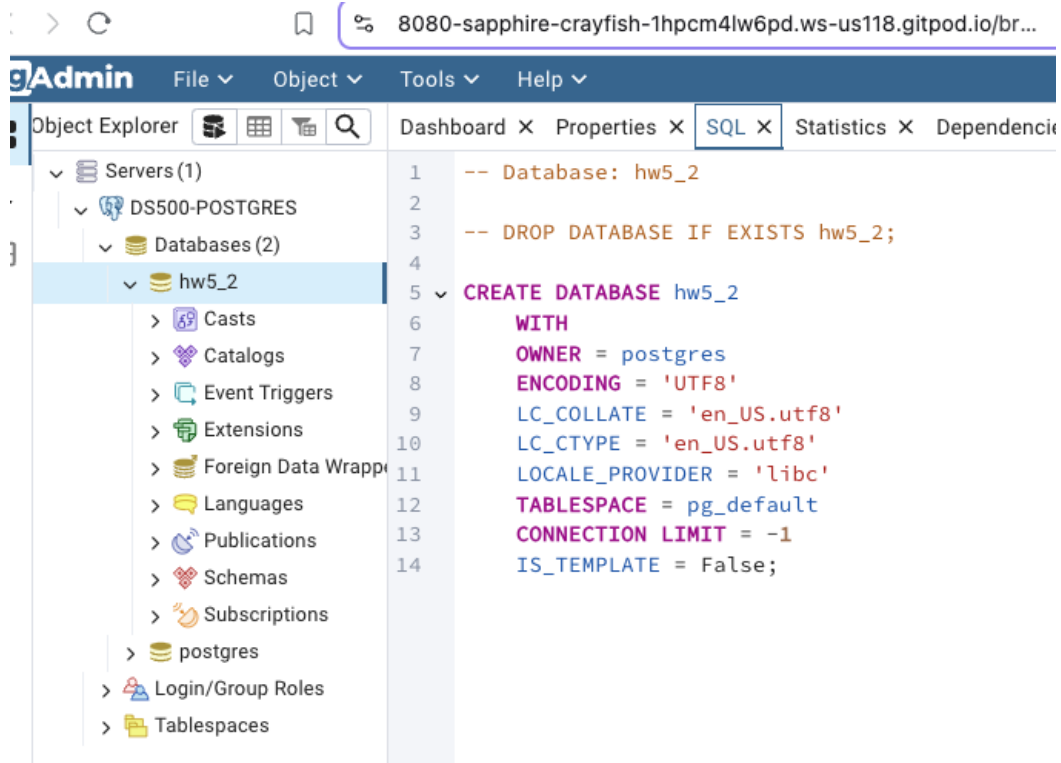
At the bottom of the dialog, there are three buttons: 'Close' (with an X icon), 'Reset' (with a circular arrow icon), and 'Save' (with a floppy disk icon). There are also two small icons (an 'i' and a '?') on the left side of the bottom bar.

5. Note that database 'flight' is already there.

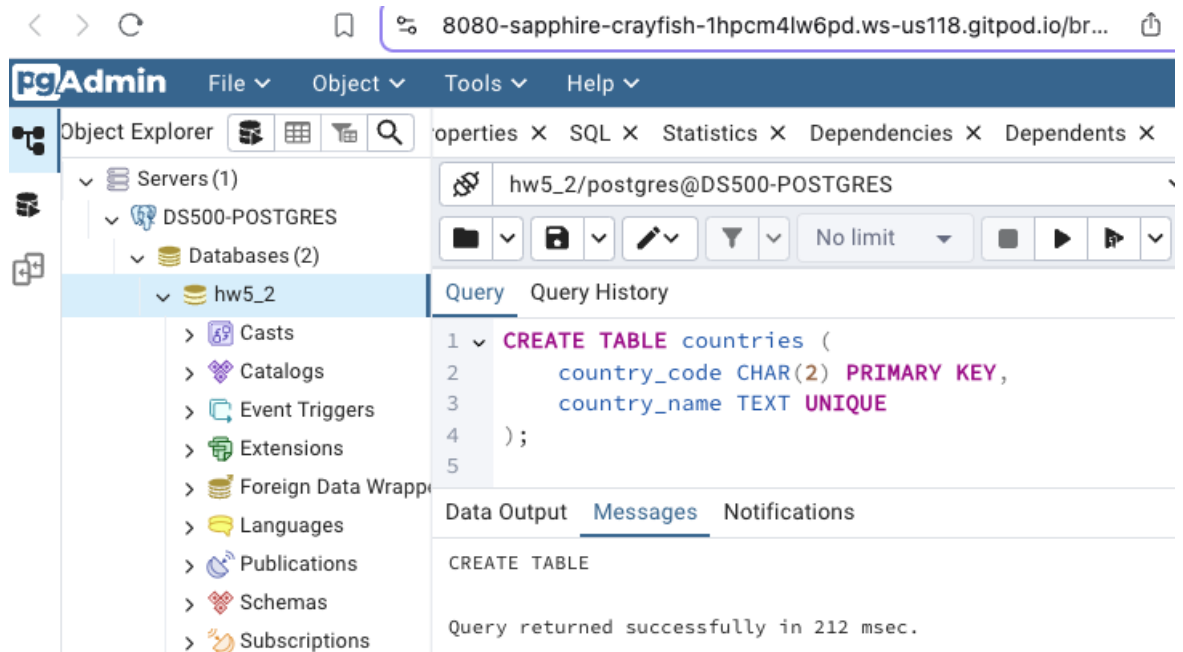
The screenshot displays the pgAdmin 4 interface. On the left, the Object Explorer shows a tree structure: Servers (1) > bd_postgres > Databases (2) > flight. Under the 'flight' database, various objects like Casts, Catalogs, Event Triggers, Extensions, Foreign Data Wrappers, Languages, Publications, and Schemas (1) are listed. The 'public' schema is expanded, showing objects like Aggregates, Collations, Domains, FTS Configurations, FTS Dictionaries, FTS Parsers, FTS Templates, Foreign Tables, Functions, Materialized Views, Operators, Procedures, Sequences, and Tables (1). The 'faa_airport_stats' table is selected and highlighted. On the right, the SQL editor shows the following SQL script:

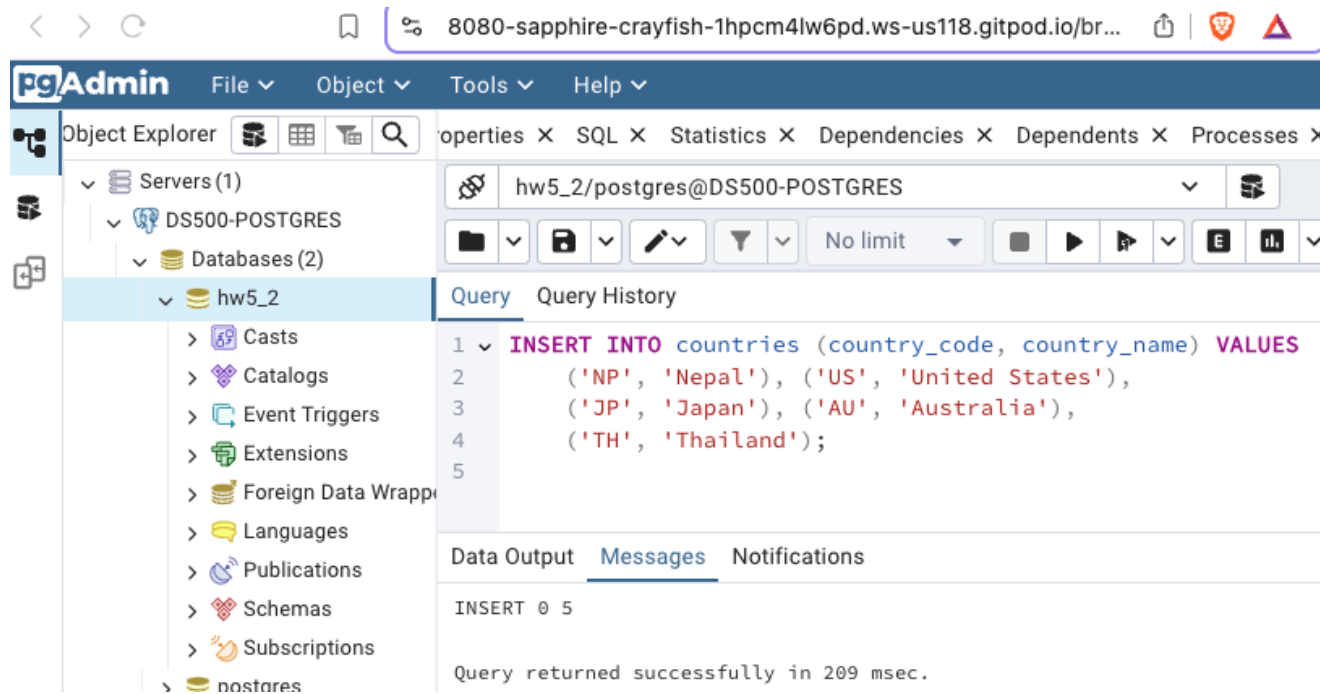
```
1  -- Table: public.faa_airport_stats
2
3  -- DROP TABLE IF EXISTS public.faa_airport_stats;
4
5  CREATE TABLE IF NOT EXISTS public.faa_airport_stats
6  (
7      id integer NOT NULL,
8      passenger_count integer,
9      quarter text COLLATE pg_catalog."default",
10     avg_comp_airfare numeric(6,2),
11     gdp numeric(12,2),
12     q1 numeric(6,2),
13     q2 numeric(6,2),
14     q3 numeric(6,2),
15     q_y numeric(6,2),
16     covid_19_ text COLLATE pg_catalog."default",
17     net_income_fin_position_ numeric(10,2),
18     weather_natural_disaster text COLLATE pg_catalog."default",
19     avg_airfare numeric(6,2),
20     year integer,
21     entry_exit_in_market text COLLATE pg_catalog."default",
22     recession boolean,
23     price_alt_transport_gasoline_ numeric(6,2),
24     CONSTRAINT faa_airport_stats_pkey PRIMARY KEY (id)
25 )
26
27 TABLESPACE pg_default;
28
29 ALTER TABLE IF EXISTS public.faa_airport_stats
30     OWNER to postgres;
```

OR create a new database say 'hw5_2'



6. Create new table countries and insert values.





5. Use Chisel to tunnel gitpod and COlab

1. Use chisel tunnel to connect to the GitPod postgresQL server container from Google Colab, and verify that you can execute SQL queries to pull events and their corresponding location information into a pandas dataframe.

1.1 Run the following in the gitpod terminal (Chiesel server)

```
docker run --name postgres_container -e POSTGRES_PASSWORD=12345 -e POSTGRES_USER=postgres -d -p 5432:5432 -v /workspace/empty/pg_files:/var/lib/postgresql/data postgres
```

```
curl https://i.jpillora.com/chisel! | bash
```

```
chisel server --port 5433
```

Port	Address	Protocol	Description	State
5432	https://5432-gitpodio-empty-zwk6koq0okm.ws-us118.gitpod.io	HTTP		open (public)
5433	https://5433-gitpodio-empty-zwk6koq0okm.ws-us118.gitpod.io	HTTP		open (public)
43319	https://43319-gitpodio-empty-zwk6koq0okm.ws-us118.gitpod.io	HTTP		open (private)

1.2 Run the following in the Google Colab (CHiesel CLinet)

```
!curl https://i.jpillora.com/chisel! | bash
```

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload Upload	Total	Spent	Left	Speed
100	5752	0 5752	0 0	7620	0	--:--:--	7628

Installing jpillora/chisel latest (linux/amd64).....
100.0%
Installed at /usr/local/bin/chisel

```
[ ] !kill chisel # Kill any old client
```

```
!nohup chisel client https://5433-gitpodio-empty-zwk6koq0okm.ws-us118.gitpod.io 0.0.0.0:5438:5432 &
```

nohup: appending output to 'nohup.out'

```
[14] import time  
time.sleep(3)  
!cat nohup.out
```

```
2025/07/05 15:17:10 client: Connecting to wss://5433-gitpodio-empty-10s8yiot1l  
2025/07/05 15:17:10 client: tun: proxy#5438=>0.0.0.0:5432: Listening  
2025/07/05 15:17:10 client: Connected (Latency 1.152315ms)
```

postgres=# create database gu_mit;

postgres=# \c gu_mit

You are now connected to database "gu_mit" as user "postgres".

gu_mit=# \dt

Did not find any relations.

```
gu_mit=# CREATE TABLE students (  
  student_id SERIAL PRIMARY KEY,  
  full_name VARCHAR(100),  
  email VARCHAR(100) UNIQUE,  
  enrollment_year INTEGER,
```

```
major VARCHAR(50),  
gpa NUMERIC(3,2)  
);
```

```
INSERT INTO students (full_name, email, enrollment_year, major, gpa) VALUES
```

```
('Sita KC', 'sita@gu.edu', 2022, 'Computer Science', 3.85),('Ram Patel', 'ram@gu.com', 2022, 'Environment Engineering',  
3.77);
```